R13

Code No: 115AP

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, November/December - 2018
COMPILER DESIGN
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a) Define Boot strapping? [2]
b) What are the draw backs of predictive parsing? [3]
c) What are the actions performed by Shift reduce parser? [2]
d) Differentiate between SLR, LALR and CLR parsers. [3]
e) What are the applications of DAG? [2]
f) What are the advantages of stack storage allocation strategy? [3]
g) Define Basic block. What are the rules for defining a basic block? [2]
h) What is common sub expression elimination? [3]
i) Define Dead code elimination? [2]
j) What is register allocation? Give a brief description. [3]

PART - B

(50 Marks)

2.a) Define Regular Expression. Explain about the Properties of Regular Expressions.
b) Differentiate between Top down and bottom up parsing techniques. [5+5]

OR

3.a) Define Compiler? Explain in brief about the types of lexical errors with example.
b) Construct a Predictive parsing table for the Grammar [4+6]
E→E+T/T, T→T*F/F, F→ (E)/id?

4.a) Construct LALR Parsing table for the grammar
S→L=R/R
L→*R/id
R→L
b) Define Ambiguous Grammar? Check whether the grammar
S→aAB,
A→bC/cd,
C→cd,
B→c/d Is Ambiguous or not? [6+4]

OR

5.a) Construct SLR Parsing table for the grammar
S→ (L)|a
L→L,s|s
b) Discuss in brief about model of LR parser. [6+4]

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6.a) Construct a Quadruple, Triple and Indirect Triple for the statement a+a*(b-c) +(b-c)*d?

b) Differentiate between Static and Dynamic Storage allocation Strategies. [5+5]

OR

7.a) Construct an annotated parse tree for real id1, id2, id3.

b) Explain in brief about equivalence of type expressions. [5+5]

8.a) Explain in brief about different Principal sources of optimization techniques with suitable examples.

b) Define Flow Graph? Explain how a given program can be converted into flow graph? [5+5]

OR

9.a) Explain in brief about function preserving transformations on basic blocks.

b) Explain in brief about optimization of basic blocks. [5+5]

10.a) Explain in detail about global common sub expression elimination technique.

b) Explain in brief about Induction variable elimination. [5+5]

OR

11.a) Explain in brief about the issues in the design of code generator.

b) Explain in detail about peep hole optimization. [5+5]

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